ALGEBRA 1 · PERSONALIZED PRACTICE

# Math Practice

2026-11-13 · 20 questions

## I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

**1.** I can identify the domain and range of a relation.

**List the range.**

$(2, 5), (4, 7), (6, 5)$

**2.** I can identify the domain and range of a relation.

For $f(x) = x + 2$ with inputs $\{-1, 0, 3\}$, list its outputs. Then extend 4, 9, 14, … one term.

**3.** I can identify the domain and range of a relation.

The function $f(x) = \sqrt{x}$ has real outputs. Explain why negative inputs are excluded.

**4.** I can identify the domain and range of a relation.

Find the range of $\{(1, 4), (2, 4), (3, -1)\}$.

## I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

**5.** I can connect linear growth in tables, graphs, and equations.

**Complete the table.**

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| y | 4 | 7 | 10 | ? |

**6.** I can connect linear growth in tables, graphs, and equations.

The outputs 2, 5, 8, 11 increase by a constant amount. What is the change per step?

**7.** I can connect linear growth in tables, graphs, and equations.

A table has $x = 0, 1, 2, 3$ and $y = 3, 7, 11, 15$. Write a rule.

**8.** I can connect linear growth in tables, graphs, and equations.

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| A | 1 | 4 | 7 | 10 |
| B | 1 | 4 | 9 | 16 |

Which is linear? Give numerical evidence.

## I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

**9.** I can calculate and interpret slope as a constant rate of change.

As $x$ rises by 4, $y$ falls by 12. Find the slope.

**10.** I can calculate and interpret slope as a constant rate of change.

**Find the run.**

$(-1,8)$ to $(6,10)$

**11.** I can calculate and interpret slope as a constant rate of change.

Find the slope through (0,3) and (4,11). Then evaluate $f(2)$ for $f(x) = 3x - 1$.

**12.** I can calculate and interpret slope as a constant rate of change.

**Find the slope.**

(−2,1), (2,9)

## I CAN COMPARE VERTICAL AND HORIZONTAL CHANGES TO FIND SLOPE.

**13.** I can compare vertical and horizontal changes to find slope.

**Find the slope.**

Rise −6, run 3

**14.** I can compare vertical and horizontal changes to find slope.

Find rise/run from (−2,1) to (2,−3). Then find $y$ when $x = 4$ in $y = 2x + 1$.

**15.** I can compare vertical and horizontal changes to find slope.

Find the slope between $(-2, 5)$ and $(2, -3)$.

**16.** I can compare vertical and horizontal changes to find slope.

Find $\Delta y$ and $\Delta x$ from $(1, 3)$ to $(4, 9)$.

## I CAN WRITE AND INTERPRET LINEAR EQUATIONS IN SLOPE-INTERCEPT FORM.

**17.** I can write and interpret linear equations in slope-intercept form.

Write an equation with slope 2 and intercept −5. Decide if $(0, 1), (1, 4), (0, 2)$ is a function.

**18.** I can write and interpret linear equations in slope-intercept form.

**Find the slope.**

$y = -\frac{1}{2}x + 6$

**19.** I can write and interpret linear equations in slope-intercept form.

**Find the y-intercept.**

$y = 5x - 7$

**20.** I can write and interpret linear equations in slope-intercept form.

**Write an equation.**

Slope $-3$, intercept $4$.